Chapter 2: Conditional Probability and Bayes' Theorem

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CSAI2121 Probability and Statistics

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Learning Objectives

By the end of this lecture, you should be able to:

- Define and compute conditional probability
- Apply the multiplication rule
- Use probability trees for multi-stage problems
- Understand and apply the Law of Total Probability
- Derive and apply Bayes' Theorem
- Interpret *prior*, *likelihood*, and *posterior* in classification contexts

Conditional Probability

Motivation: Why Conditional Probability?

- Last week, we treated events as isolated, static objects.
- In reality, information **updates** our beliefs.
- Example: *"What is the probability an email is spam, given that it contains the word 'free'?"*
- This idea is the foundation of:
  - Spam filters
  - Medical diagnosis systems
  - Naive Bayes classifiers

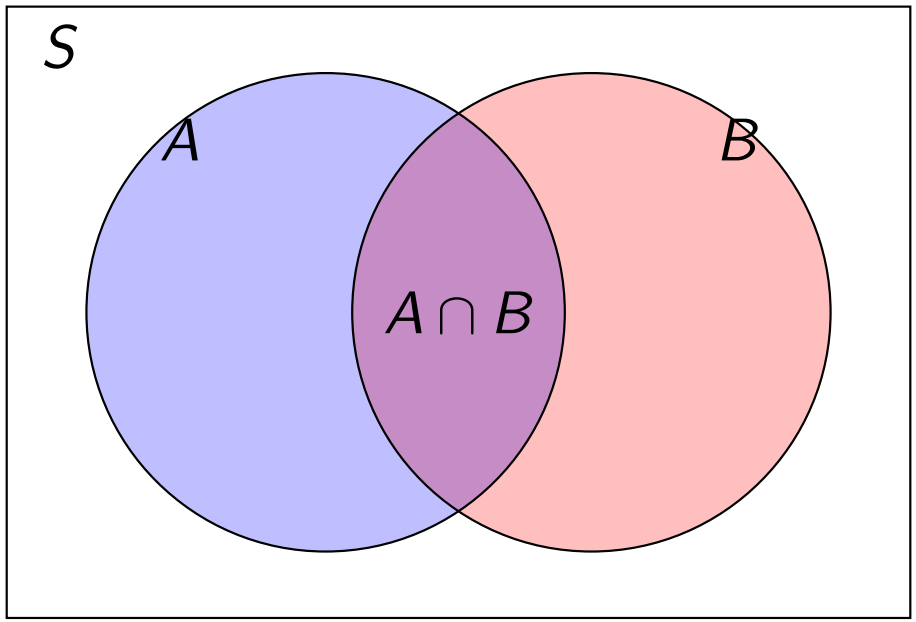
Definition of Conditional Probability

Definition

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0$$

- Read as: "the probability of A given B"
- Conditioning on B **restricts** the sample space to B
- Intuition: we already know B happened — how does that change the likelihood of A?

Visual Intuition (Venn Diagram)



- Conditioning on B means restricting attention to the region B.
- Within B, only $A \cap B$ is where A *also* occurs.
- $P(A \mid B)$ = fraction of B that is covered by $A \cap B$.

Worked Example 1: Dice

Roll a fair six-sided die.

- $A =$ "the roll is even" $= \{2, 4, 6\}$
- $B =$ "the roll is at least 4" $= \{4, 5, 6\}$

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$$P(B) = \frac{3}{6}$$

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$$P(B) = \frac{3}{6}$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{3/6} = \frac{2}{3}$$

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$$P(B) = \frac{3}{6}$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{3/6} = \frac{2}{3}$$

Compare: $P(A) = \frac{1}{2}$, but $P(A \mid B) = \frac{2}{3}$ — conditioning **changed** the probability.

Worked Example 2: Dataset Filtering (AI Context)

A dataset of 200 emails is classified as follows:

	Contains "free"	No "free"	Total
Spam	60	20	80
Not Spam	20	100	120
Total	80	120	200

Worked Example 2: Dataset Filtering (AI Context)

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	Contains "free"	No "free"	Total
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$$P(\text{Spam} \mid \text{"free"}) = \frac{P(\text{Spam} \cap \text{"free"})}{P(\text{"free"})} = \frac{60/200}{80/200} = \frac{60}{80} = 0.75$$

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	Contains "free"	No "free"	Total
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$$P(\text{Spam} \mid \text{"free"}) = \frac{P(\text{Spam} \cap \text{"free"})}{P(\text{"free"})} = \frac{60/200}{80/200} = \frac{60}{80} = 0.75$$

Conditioning on a feature ("free") is exactly what a spam filter does when scoring an email.

The Multiplication Rule

Rearranging the definition of conditional probability in both directions:

$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A)$$

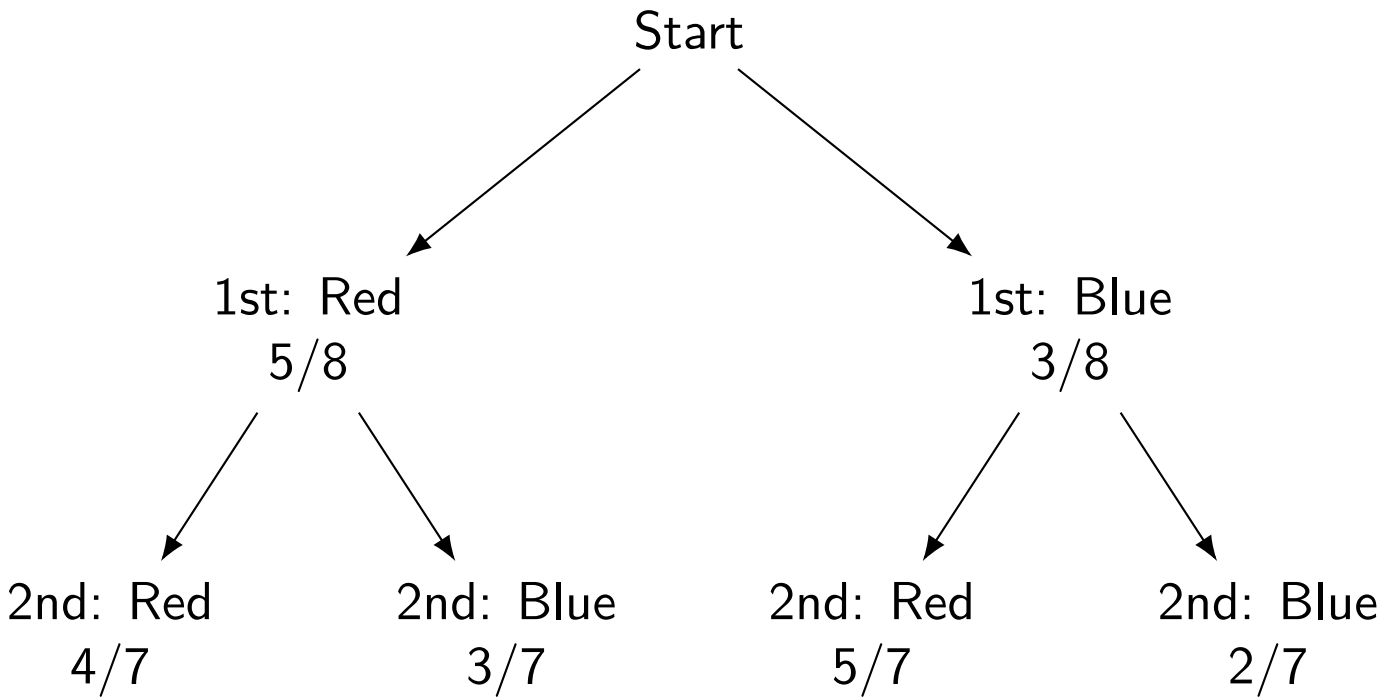
- Useful when the joint probability $P(A \cap B)$ is unknown,
- but a conditional probability and a marginal probability **are** known.
- This is the algebraic engine behind probability trees and Bayes' Theorem.

Multi-Stage Experiments: Probability Trees

- A **probability tree** represents a sequence of random stages.
- Each branch is labeled with a conditional probability, given everything above it.
- The probability of a full path = product of branch probabilities (multiplication rule).
- Very useful for problems with 2–3 sequential stages.

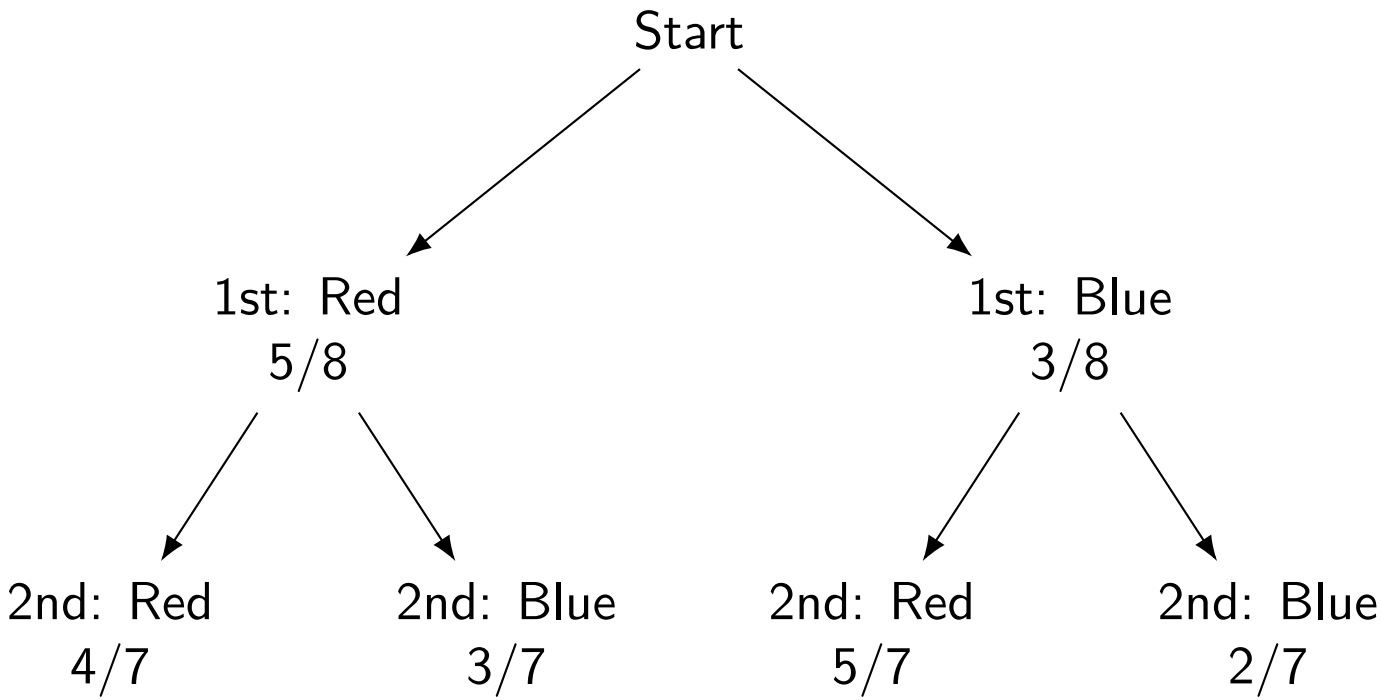
Worked Example 3: Sequential Draws

An urn contains 5 red and 3 blue balls (8 total). Draw **twice without replacement**.



Worked Example 3: Sequential Draws

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$$P(\text{2nd} = \text{Red}) = \frac{5}{8} \cdot \frac{4}{7} + \frac{3}{8} \cdot \frac{5}{7} = \frac{20}{56} + \frac{15}{56} = \frac{35}{56} = \frac{5}{8}$$

Interesting: the unconditional probability that the *2nd* draw is red equals the original proportion 5/8.

Class Exercise (Pair Work)

Problem

A box contains 4 defective and 16 working chips (20 total). Two chips are drawn **without replacement**.

1. Draw the probability tree for the first two draws.
2. Compute $P(\text{2nd chip is defective})$.
3. Compute $P(\text{both chips are defective})$.

Work in pairs (5 minutes), then we discuss as a class.

Instructor Demo 1: Conditional Probability with Cards

A standard 52-card deck. One card is drawn at random.

- $A =$ "card is a King"
- $B =$ "card is a face card" (King, Queen, or Jack)

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Face cards: $4 + 4 + 4 = 12$, so $P(B) = \dfrac{12}{52}$

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Every King is a face card, so $A \cap B = A$, and $P(A \cap B) = \dfrac{4}{52}$

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Every King is a face card, so $A \cap B = A$, and $P(A \cap B) = \dfrac{4}{52}$

$$P(A \mid B) = \frac{4/52}{12/52} = \frac{4}{12} = \frac{1}{3}$$

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Every King is a face card, so $A \cap B = A$, and $P(A \cap B) = \dfrac{4}{52}$

$$P(A \mid B) = \frac{4/52}{12/52} = \frac{4}{12} = \frac{1}{3}$$

Compare to $P(A) = \dfrac{4}{52} = \dfrac{1}{13}$. Knowing the card is a face card makes it much more likely to be a King ($1/13 \to 1/3$).

Instructor Demo 2: Multiplication Rule in Quality Control

A component passes through two inspection stages.

- $P(\text{Pass Stage 1}) = 0.90$
- $P(\text{Pass Stage 2} \mid \text{Pass Stage 1}) = 0.80$

Question: What is the probability a component passes *both* stages?

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Question: What is the probability a component passes *both* stages?

$$P(\text{Pass 1} \cap \text{Pass 2}) = P(\text{Pass 2} \mid \text{Pass 1})\, P(\text{Pass 1}) = (0.80)(0.90) = 0.72$$

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72% of components make it through the full pipeline.

This two-stage structure mirrors a two-stage ML pipeline, e.g., a fast filter followed by a slower, more accurate classifier.

Independence

Recap: Independence

- Last week: two events A and B are **independent** if knowing one gives **no information** about the other.
- Today, we can state this precisely using conditional probability.

Formal Definition via Conditional Probability

Independence

A and B are independent if and only if

$$P(A \mid B) = P(A) \quad \Longleftrightarrow \quad P(A \cap B) = P(A)\,P(B)$$

- This is a **stronger**, more usable definition than last week's informal statement.
- Conditioning on B does not change the probability of A at all.

Independence vs. Mutually Exclusive

These are different concepts!

- Mutually exclusive: $P(A \cap B) = 0$
- Independent: $P(A \cap B) = P(A)P(B)$

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Key Point

If A and B are mutually exclusive *and both have nonzero probability*, they are **NOT** independent.

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Key Point

If A and B are mutually exclusive *and both have nonzero probability*, they are **NOT** independent.

Quick check: suppose $P(A) = 0.3$, $P(B) = 0.4$, and $A \cap B = \varnothing$.

$$P(A \cap B) = 0 \quad \text{but} \quad P(A)P(B) = 0.12 \neq 0$$

So A, B mutually exclusive $\Rightarrow$ dependent (in this case).

Worked Example 4: Checking Independence

Joint probabilities for "Rain" (A) and "Heavy Traffic" (B):

	Heavy Traffic	Light Traffic
Rain	0.15	0.15
No Rain	0.35	0.35

Worked Example 4: Checking Independence

Joint probabilities for "Rain" (A) and "Heavy Traffic" (B):

	Heavy Traffic	Light Traffic
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$$P(\text{Rain}) = 0.30, \qquad P(\text{Heavy}) = 0.50$$

Worked Example 4: Checking Independence

Joint probabilities for "Rain" (A) and "Heavy Traffic" (B):

	Heavy Traffic	Light Traffic
Rain	0.15	0.15
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$$P(\text{Rain}) = 0.30, \qquad P(\text{Heavy}) = 0.50$$

$$P(\text{Rain}) \times P(\text{Heavy}) = 0.30 \times 0.50 = 0.15 = P(\text{Rain} \cap \text{Heavy})$$

Worked Example 4: Checking Independence

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$$P(\text{Rain}) = 0.30, \qquad P(\text{Heavy}) = 0.50$$

$$P(\text{Rain}) \times P(\text{Heavy}) = 0.30 \times 0.50 = 0.15 = P(\text{Rain} \cap \text{Heavy})$$

Since the equality holds, **Rain and Heavy Traffic are independent** in this (constructed) dataset.

Independence in AI/ML Context

- **Naive Bayes** classifiers assume that features are *conditionally independent* given the class label.
- This assumption is almost always technically false — but it drastically simplifies computation, and often works well in practice.
- Example: assuming the words "free" and "win" occur independently given "spam", even though real text has correlations.

We will return to this in detail when we reach Bayes' Theorem later today.

Class Exercise

Problem

Given the following joint probability table, determine whether A and B are independent.

	B	B^c
A	0.10	0.30
A^c	0.20	0.40

Hint: Compute $P(A)$, $P(B)$, and compare $P(A)P(B)$ with $P(A \cap B)$.

Instructor Demo 3: Independence Check — Lifestyle Survey

A survey of 1000 adults reports:

$$P(\text{Smoker}) = 0.30, \qquad P(\text{Exercises Regularly}) = 0.40,$$

$$P(\text{Smoker} \cap \text{Exercises}) = 0.10$$

Instructor Demo 3: Independence Check — Lifestyle Survey

A survey of 1000 adults reports:

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If independent, we would expect:

$$P(\text{Smoker})\,P(\text{Exercises}) = (0.30)(0.40) = 0.12$$

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But the actual joint probability is $0.10 \neq 0.12$.

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But the actual joint probability is $0.10 \neq 0.12$.

Conclusion

Smoking and Exercise are **NOT independent** in this survey — smokers exercise somewhat less than we would expect under independence.

Instructor Demo 4: Independence Check for Naive Bayes Features

In an image dataset:

$$P(\text{“sky”}) = 0.6, \qquad P(\text{“cloud”}) = 0.3, \qquad P(\text{“sky”} \cap \text{“cloud”}) = 0.18$$

Instructor Demo 4: Independence Check for Naive Bayes Features

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Check:

$$P(\text{“sky”})\,P(\text{“cloud”}) = (0.6)(0.3) = 0.18 = P(\text{“sky”} \cap \text{“cloud”})$$

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The equality holds exactly — these two features **are independent** in this dataset, exactly the assumption Naive Bayes relies on.

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The equality holds exactly — these two features **are independent** in this dataset, exactly the assumption Naive Bayes relies on. *In real datasets, features are rarely perfectly independent — but Naive Bayes often still performs well even when the assumption only holds approximately.*

Law of Total Probability

Motivation: Law of Total Probability

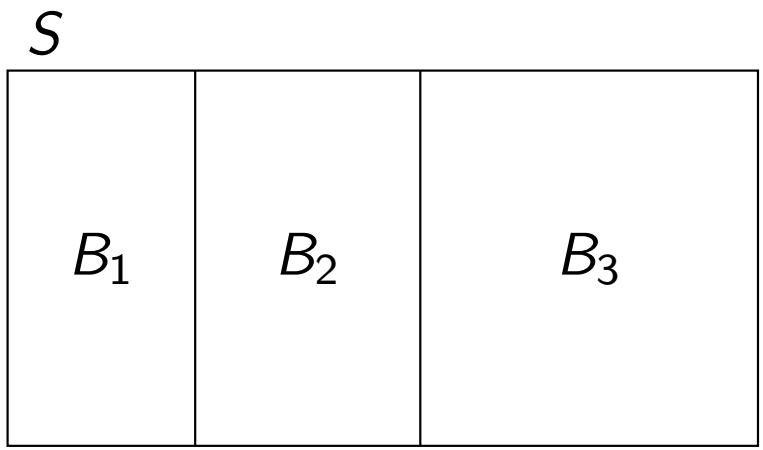
- Sometimes we cannot compute $P(A)$ directly.
- But we **can** break the sample space into known pieces (scenarios).
- Example: What is the overall defect rate of a product, if it is manufactured at three different factories with different defect rates?

Partitions of the Sample Space

Definition

Events $B_1, B_2, \ldots, B_n$ form a **partition** of the sample space S if:

1. They are **mutually exclusive**: $B_i \cap B_j = \varnothing$ for $i \neq j$
2. They are **exhaustive**: $B_1 \cup B_2 \cup \cdots \cup B_n = S$



Think of the sample space sliced into non-overlapping pieces.

Law of Total Probability

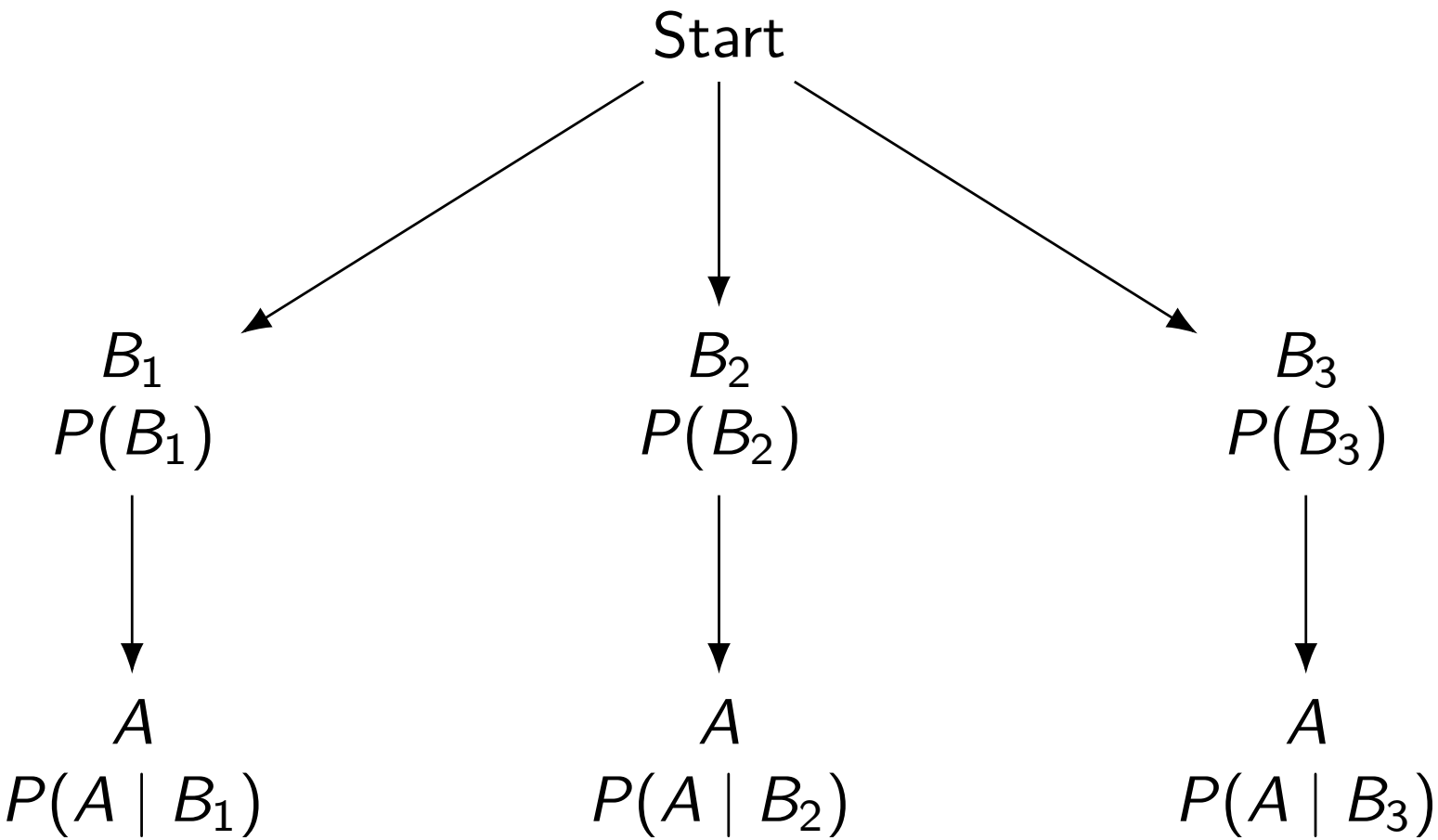
Statement

If $B_1, \ldots, B_n$ form a partition of S, then for any event A:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)\, P(B_i)$$

- Intuition: $P(A)$ is a **weighted average** of the conditional probabilities $P(A \mid B_i)$,
- weighted by how likely each scenario B_i is.

Visual: Probability Tree Connection



$P(A)$ = sum over **all paths** in the tree that end in A — exactly the Law of Total Probability.

Worked Example 5: Three Factories

A product is manufactured at three factories:

Factory	Share of Output	Defect Rate
A	30%	2%
B	50%	3%
C	20%	5%

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Factory	Share of Output	Defect Rate
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$$P(\text{Defective}) = (0.30)(0.02) + (0.50)(0.03) + (0.20)(0.05)$$

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A	30%	2%
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$$P(\text{Defective}) = (0.30)(0.02) + (0.50)(0.03) + (0.20)(0.05)$$

$$= 0.006 + 0.015 + 0.010 = 0.031$$

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$$P(\text{Defective}) = (0.30)(0.02) + (0.50)(0.03) + (0.20)(0.05)$$

$$= 0.006 + 0.015 + 0.010 = 0.031$$

The overall defect rate across all factories is **3.1%**.

Worked Example 6: AI Context — Model Ensemble

An input-routing system sends queries to one of three models:

Model	Fraction of Inputs	Error Rate
Model 1	40%	5%
Model 2	35%	8%
Model 3	25%	3%

Worked Example 6: AI Context — Model Ensemble

An input-routing system sends queries to one of three models:

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$$P(\text{Error}) = (0.40)(0.05) + (0.35)(0.08) + (0.25)(0.03)$$

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Model 1	40%	5%
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$$P(\text{Error}) = (0.40)(0.05) + (0.35)(0.08) + (0.25)(0.03)$$

$$= 0.020 + 0.028 + 0.0075 = 0.0555$$

Worked Example 6: AI Context — Model Ensemble

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Model 1	40%	5%
Model 2	35%	8%
Model 3	25%	3%

$$P(\text{Error}) = (0.40)(0.05) + (0.35)(0.08) + (0.25)(0.03)$$

$$= 0.020 + 0.028 + 0.0075 = 0.0555$$

The overall system error rate is **5.55%**, even though no single model has that exact error rate.

Class Exercise

Problem

A hospital receives patients from three clinics:

- Clinic 1: 25% of patients, 4% require surgery
- Clinic 2: 45% of patients, 2% require surgery
- Clinic 3: 30% of patients, 6% require surgery

Compute the overall probability that a randomly selected patient requires surgery.

Instructor Demo 5: Shipping Delays Across Warehouses

An online store ships from 3 warehouses:

Warehouse	Share of Shipments	Late Rate
W1	40%	5%
W2	35%	2%
W3	25%	8%

Instructor Demo 5: Shipping Delays Across Warehouses

An online store ships from 3 warehouses:

Warehouse	Share of Shipments	Late Rate
W1	40%	5%
W2	35%	2%
W3	25%	8%

$$P(\text{Late}) = (0.40)(0.05) + (0.35)(0.02) + (0.25)(0.08)$$

Instructor Demo 5: Shipping Delays Across Warehouses

An online store ships from 3 warehouses:

Warehouse	Share of Shipments	Late Rate
W1	40%	5%
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W3	25%	8%

$$P(\text{Late}) = (0.40)(0.05) + (0.35)(0.02) + (0.25)(0.08)$$

$$= 0.020 + 0.007 + 0.020 = 0.047$$

Instructor Demo 5: Shipping Delays Across Warehouses

An online store ships from 3 warehouses:

Warehouse	Share of Shipments	Late Rate
W1	40%	5%
W2	35%	2%
W3	25%	8%

$$P(\text{Late}) = (0.40)(0.05) + (0.35)(0.02) + (0.25)(0.08)$$

$$= 0.020 + 0.007 + 0.020 = 0.047$$

Overall, **4.7%** of shipments will be late.

Instructor Demo 6: Overall Click-Through Rate (AI Context)

A recommender system randomly selects one of three algorithms for each user:

Algorithm	Selection Probability	Click-Through Rate
Algorithm 1	50%	10%
Algorithm 2	30%	15%
Algorithm 3	20%	5%

Instructor Demo 6: Overall Click-Through Rate (AI Context)

A recommender system randomly selects one of three algorithms for each user:

Algorithm	Selection Probability	Click-Through Rate
Algorithm 1	50%	10%
Algorithm 2	30%	15%
Algorithm 3	20%	5%

$$P(\text{Click}) = (0.50)(0.10) + (0.30)(0.15) + (0.20)(0.05)$$

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Algorithm	Selection Probability	Click-Through Rate
Algorithm 1	50%	10%
Algorithm 2	30%	15%
Algorithm 3	20%	5%

$$P(\text{Click}) = (0.50)(0.10) + (0.30)(0.15) + (0.20)(0.05)$$

$$= 0.050 + 0.045 + 0.010 = 0.105$$

Instructor Demo 6: Overall Click-Through Rate (AI Context)

A recommender system randomly selects one of three algorithms for each user:

Algorithm	Selection Probability	Click-Through Rate
Algorithm 1	50%	10%
Algorithm 2	30%	15%
Algorithm 3	20%	5%

$$P(\text{Click}) = (0.50)(0.10) + (0.30)(0.15) + (0.20)(0.05)$$

$$= 0.050 + 0.045 + 0.010 = 0.105$$

The overall system-wide click-through rate is **10.5%**.

Bayes' Theorem

Motivation: Reversing the Condition

- We often know $P(\text{evidence} \mid \text{cause})$,
- but we actually want $P(\text{cause} \mid \text{evidence})$.

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- We often know $P(\text{evidence} \mid \text{cause})$,
- but we actually want $P(\text{cause} \mid \text{evidence})$.

Example: A clinical trial tells us $P(\text{positive test} \mid \text{disease})$.
But a patient who tests positive wants to know $P(\text{disease} \mid \text{positive test})$.

Motivation: Reversing the Condition

- We often know $P(\text{evidence} \mid \text{cause})$,
- but we actually want $P(\text{cause} \mid \text{evidence})$.

Example: A clinical trial tells us $P(\text{positive test} \mid \text{disease})$.
But a patient who tests positive wants to know $P(\text{disease} \mid \text{positive test})$.

This reversal is exactly what Bayes' Theorem provides.

Deriving Bayes' Theorem

Start from the multiplication rule, written in both directions:

$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A)$$

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Divide both sides by $P(B)$:

Bayes' Theorem

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}$$

This is simple algebra — no calculus required.

Bayes' Theorem (General Form)

When there are several competing hypotheses $A_1, \ldots, A_n$ (a partition), combine with the Law of Total Probability:

Bayes' Theorem, General Form

$$P(A_i \mid B) = \frac{P(B \mid A_i)\, P(A_i)}{\sum_j P(B \mid A_j)\, P(A_j)}$$

- The denominator is just the Law of Total Probability from Segment C.
- This shows exactly how today's two big ideas connect.

Naming the Terms

- $P(A)$ = **Prior**
  belief before seeing evidence
- $P(B \mid A)$ = **Likelihood**
  how probable the evidence is, under hypothesis A
- $P(B)$ = **Evidence**
  total probability of the observed data
- $P(A \mid B)$ = **Posterior**
  updated belief after seeing evidence

This vocabulary will reappear extensively in MLE and MAP (Weeks 8 and 10).

Worked Example 7: Medical Testing (Classic)

- Disease prevalence: $P(\text{Disease}) = 0.01$
- Test sensitivity: $P(\text{Positive} \mid \text{Disease}) = 0.95$
- False positive rate: $P(\text{Positive} \mid \text{No Disease}) = 0.05$

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$$= \frac{0.0095}{0.0095 + 0.0495} = \frac{0.0095}{0.0590} \approx 0.161$$

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Surprising Result

Even with a positive test, the probability of actually having the disease is only about **16%**!

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Compare with the same test, but disease prevalence $= 0.50$ instead of 0.01:

$$P(\text{Disease} \mid \text{Positive}) = \frac{(0.95)(0.50)}{(0.95)(0.50) + (0.05)(0.50)} = \frac{0.475}{0.500} = 0.95$$

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Key Lesson

The same test can give a very different posterior depending on the **prior**.
The prior matters enormously — this is the *base-rate fallacy*.

Worked Example 8: Spam Classification (AI Context)

- $P(\text{Spam}) = 0.4 \quad (\text{prior})$
- $P(\text{“free”} \mid \text{Spam}) = 0.6$
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Total probability of seeing “free”:

$$P(\text{“free”}) = (0.6)(0.4) + (0.1)(0.6) = 0.24 + 0.06 = 0.30$$

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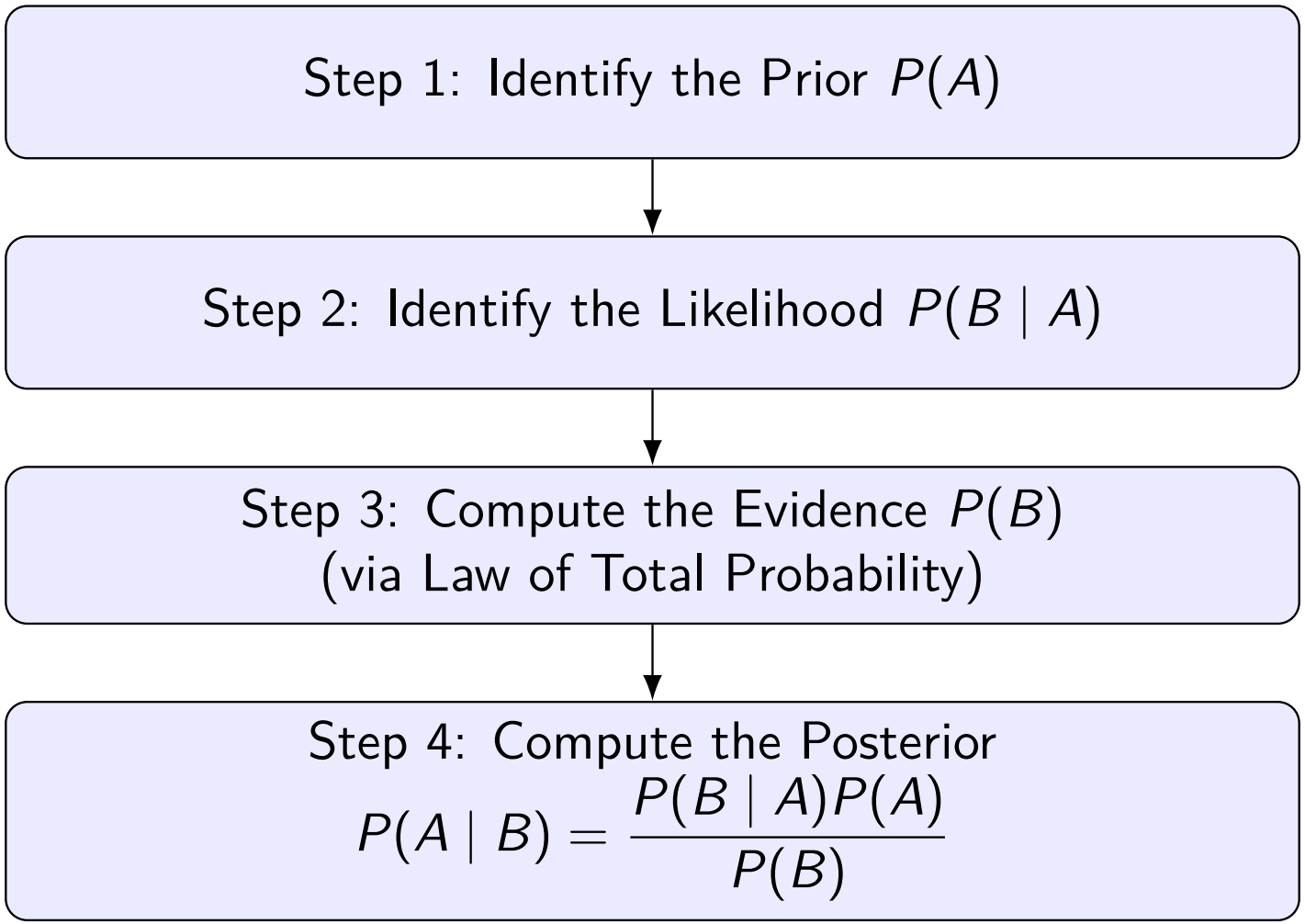
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This is a direct preview of how a **Naive Bayes** spam classifier works.

Visual Summary: The Bayes' Theorem "Recipe"



Use this template for *any* Bayes' Theorem problem.

Common Mistakes with Bayes' Theorem

- **Transposed conditional fallacy**: confusing $P(A \mid B)$ with $P(B \mid A)$
  - e.g., "95% sensitivity" does *not* mean "95% of positive tests indicate disease"
- Forgetting to use the **full denominator** (Law of Total Probability) when there are multiple hypotheses
- Forgetting the **prior** entirely — treating the likelihood as if it were the answer

Instructor Demo 7: Which Die Is It?

Two dice: one **fair**, one **loaded** so that $P(6) = 0.5$ and each of 1–5 has probability 0.1.
You pick one die at random (50/50) and roll a **6**. Find
$P(\text{Loaded} \mid \text{Rolled a 6})$.

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$$P(6 \mid \text{Fair}) = \frac{1}{6} \approx 0.167, \qquad P(6 \mid \text{Loaded}) = 0.5$$

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$$P(6) = (0.5)(0.167) + (0.5)(0.5) = 0.083 + 0.250 = 0.333$$

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Rolling a 6 pushes our belief that the die is loaded from a 50% prior up to a **75%** posterior.

Instructor Demo 8: Revisiting the Factories

Recall Worked Example 5: Factory A (30%, 2% defect), B (50%, 3%), C (20%, 5%), giving $P(\text{Defective}) = 0.031$.

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New question: Given that an item *is* defective, which factory most likely made it?

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New question: Given that an item *is* defective, which factory most likely made it?

$$P(A \mid \text{Def}) = \frac{0.006}{0.031} \approx 0.194, \qquad P(B \mid \text{Def}) = \frac{0.015}{0.031} \approx 0.484$$

$$P(C \mid \text{Def}) = \frac{0.010}{0.031} \approx 0.323$$

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$$P(C \mid \text{Def}) = \frac{0.010}{0.031} \approx 0.323$$

Insight

Factory B produces so much output that it is the **most likely source** of any given defect, even though its defect rate (3%) is not the highest. This completes the loop from Law of Total Probability to Bayes' Theorem.

Instructor Demo 9: Naive Bayes with Two Words

Priors: $P(\text{Spam}) = 0.5$, $P(\text{NotSpam}) = 0.5$.
Likelihoods (assumed conditionally independent given class):

$$P(\text{“win”} \mid \text{Spam}) = 0.30, \quad P(\text{“win”} \mid \text{NotSpam}) = 0.05$$

$$P(\text{“meeting”} \mid \text{Spam}) = 0.05, \quad P(\text{“meeting”} \mid \text{NotSpam}) = 0.20$$

An email contains **both** words. Classify it.

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Classified as **Spam** with 60% posterior — this is exactly how a real Naive Bayes filter combines Independence (Segment B) with Bayes' Theorem (Segment D).

Applications and Pitfalls

Bayes' Theorem in Machine Learning: Preview

- **Naive Bayes classifiers**: apply Bayes' Theorem with a (naive) conditional independence assumption among features — connects directly to Segment B.
- **Bayesian inference** in general: treat unknown parameters as random variables, update belief from prior to posterior as data arrives.
- Full **Bayesian parameter estimation** (MAP) will be covered in Week 10, once we have random variables and distributions (Weeks 3–6).

Worked Example 9 (Optional): Sequential Updating

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After 1st flip = Heads:

$$P(H) = (0.5)(0.5) + (0.5)(0.9) = 0.70$$

$$P(\text{fair} \mid H) = \frac{0.25}{0.70} \approx 0.357, \qquad P(\text{biased} \mid H) = \frac{0.45}{0.70} \approx 0.643$$

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After 2nd flip = Heads (using the updated posterior as the new prior):

$$P(\text{fair} \mid HH) \approx 0.236, \qquad P(\text{biased} \mid HH) \approx 0.764$$

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Belief in "biased" keeps increasing as more evidence accumulates — this is the essence of sequential Bayesian updating.

Class Exercise: Full Bayes Problem

Problem

A drug test has:

- Prevalence of drug use in the population: 2%
- Sensitivity (true positive rate): 90%
- False positive rate: 10%

1. Identify the prior, likelihood, and evidence.
2. Compute $P(\text{Drug Use} \mid \text{Positive Test})$.
3. Explain in one sentence why the result may seem surprising.

Summary: Key Formulas from Today

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A \mid B)\, P(B)$$

$$P(A) = \sum_i P(A \mid B_i)\, P(B_i)$$

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}$$

Connections Ahead

- Independence (today) $\rightarrow$ feature independence assumption in Naive Bayes
- Bayes' Theorem (today) $\rightarrow$ MAP estimation (Week 10)
- Prior / Likelihood / Posterior vocabulary $\rightarrow$ used throughout the rest of the course

Preview of Week 3

- Today, we treated events ("disease", "spam") as **binary outcomes**.
- Next week, we generalize this to **random variables** — numerical outcomes with full probability distributions.
- We will meet the PMF, CDF, expectation, variance, and common discrete distributions (Bernoulli, Binomial, Poisson).